



RAJALAKSHMI MADURAI

Senior QA Automation Engineer | AI Quality Engineering | Playwright |
Selenium | Java | Python | Typescript | LLM & Chatbot Testing |
Performance & Data Validation | GitHub Copilot | Dubai, UAE
|+971568897994 | rajalakshmimadurai889@gmail.com

<https://www.linkedin.com/in/rajalakshmi-madurai-033a85209/>

VISA Status : Dependent VISA with no notice period

PROFESSIONAL SUMMARY

Senior QA Automation Engineer with 6+ years of experience in Quality Engineering, Test Automation, API Testing, AI-powered application validation, and enterprise platform testing across Banking, Telecom, Cloud, and Utilities domains. Strong expertise in designing scalable automation frameworks using Playwright, Selenium, Python, JavaScript, and Java for web, API, backend, and workflow validation. Experienced in testing AI-powered systems including conversational AI, chatbot workflows, response validation, intent recognition, and backend integrations. Skilled in designing automation strategies for non-deterministic systems, regression detection, end-to-end workflow validation, CI/CD-integrated testing pipelines, and quality observability. Hands-on experience in API contract validation, backend data validation, performance testing, production release validation, and automation framework architecture using POM and reusable utilities. Strong understanding of AI quality engineering practices including adversarial testing, synthetic test data generation, prompt validation, and AI response evaluation.

CORE SKILLS

AI/LLM Testing: Conversational AI Testing, LLM Output Validation, Prompt Validation, AI Response Evaluation, AI Regression Testing, Intent & Entity Validation, Context Validation, Adversarial Testing, Synthetic Test Data Generation, AI workflow validation

Quality Engineering: Test Automation Architecture, Quality Observability, Regression Analytics, Test Strategy Design, Risk-Based Testing, Root Cause Analysis, Quality Metrics

API & Contract Testing: REST API Testing, Schema Validation, Contract Testing, Backend Validation, Microservices Testing

Performance & Reliability: JMeter, Load Testing, Latency Validation, Scalability Testing, Performance Benchmarking

Data Validation: SQL Validation, Backend Data Validation, Schema Validation, Data Integrity Validation, Production Data Verification

CI/CD & DevOps: Jenkins, GitHub Actions, GitLab, CI/CD Automation Pipelines

Security & Accessibility: Negative Testing, Security Validation, Accessibility Testing Basics

AI & Cloud Platforms: GitHub Copilot, ChatGPT, AWS SageMaker, Azure Cloud

- **Frameworks:** TestNG, LLM, Page Object Model (POM), CATS, API
- **Programming Languages:** Java, Python, JavaScript, TypeScript
- **API & Performance Testing:** Postman, Rest Assured
- **Cloud & Tools:** Azure, AWS SageMaker, Git, GitHub, GitLab
- **Test Management & Tracking:** JIRA, SQL, HP ALM, PyTest, JMeter, AI chatbot testing, Microsoft Copilot Studio, Jenkins
- **IDEs & Platforms:** Eclipse, Micro Focus UFT, Windows 10, Maven Build Integration
- **Methodologies:** Agile, Waterfall, Cross-team collaboration, Automation documentation, Process improvement, SDLC, STLC

Domains & Projects: Telecommunication – BT Group, Banking - CITI Bank, Utilities – Pacific Gas and Energy (PG&E)

PROFESSIONAL EXPERIENCE

Test Engineer | Automation Test Engineer | Tata Consultancy Services (TCS) | Chennai, India | July 2022 – Jan 2026

Automation & Process Improvement

- Designed and implemented automation frameworks using Selenium and Java to improve regression testing efficiency.
- Utilized AI-assisted coding tools such as GitHub Copilot and ChatGPT to accelerate Playwright test script development and improve automation coverage.
- • Designed automation validation strategies for AI-powered conversational workflows including intent validation, entity recognition, contextual response verification, and fallback scenario testing.
- Performed AI response quality validation for chatbot systems including response relevance, tone validation, workflow accuracy, and business-rule compliance.
- Developed reusable Playwright automation utilities and scalable testing components to improve regression stability and reduce flaky executions.
- Participated in quality engineering activities for non-deterministic systems where validation required contextual and workflow-level verification beyond traditional assertions.
- Designed edge-case and negative test scenarios for AI-driven workflows including invalid prompts, incomplete inputs, and failure recovery validation.
- Performed backend data validation and API integration testing for AI-integrated enterprise applications and workflow orchestration systems.
- Contributed to automation execution optimization within CI/CD pipelines through parallel execution strategies, reusable test data handling, and improved reporting visibility.
- Supported quality observability through defect trend analysis, regression tracking, execution reporting, and release validation dashboards.
- Collaborated with cross-functional engineering teams to define test strategies, automation scope, risk coverage, and release readiness validation for enterprise platforms. Logged and tracked defects using JIRA and HP ALM while coordinating with development and business teams for issue resolution.
- Supported production validation activities ensuring zero critical defects in banking transaction processing systems.
- Supported automation in CI/CD pipelines to ensure stable and reliable releases.
- Conversational AI testing (intent validation, entity recognition, response accuracy)
- Black-box testing of chatbot behavior from end-user perspective
- Performed performance and load validation using JMeter for enterprise applications to identify response-time bottlenecks and improve system reliability under concurrent user loads.
- Validated API response times, transaction processing latency, and backend stability during peak-volume execution scenarios.
- Collaborated with cross-functional Agile teams for sprint planning, defect triage, and release validation.
- Implemented reusable automation utilities and improved test execution stability across releases.
- Tested trade order placement, execution status validation, and reconciliation.
- Identified repetitive QA processes and implemented automated scripts to streamline test execution and reporting.

Automation Test Engineer (AWS) | Amazon Software Development Center | Chennai, India | June 2021 – June 2022

- Designed and executed automation test scripts using Selenium for AWS applications.
- Prepared detailed test case outlines and ensured complete requirements coverage.

- Developed automation scripts using Playwright with Node.js for web-based banking platforms.
- Designed scalable automation framework using Playwright Test with POM architecture.
- Reviewed business scenarios and test conditions to ensure functional and automation completeness.
- Used AWS SageMaker platforms for test data validation, workflow verification, and system analysis.
- Logged, tracked, and retested defects, ensuring proper closure and compliance with quality standards.
- Collaborated closely with developers, product owners, and QA teams in an Agile environment.
- Validated test environments and test data readiness before execution cycles.
- Managed complete defect lifecycle in JIRA including triage, RCA analysis, and release reporting dashboards.
- Reviewed business scenarios and test conditions to ensure functional and automation completeness

Technical Associate – CS Internet | Sutherland Global Services | Chennai, India | November 2016 – July 2019

- Provided technical and operational support to Amazon third-party sellers, resolving account, listing, and fulfillment issues.
- Reproduced defects and collaborated with development teams for issue resolution.
- Created and maintained test scenarios and support documentation.
- Assisted QA teams in identifying regression-prone module
- Ensured compliance with process controls and quality standards.
- Assisted automation teams by identifying regression-prone areas and supporting Selenium test execution and defect validation.
- Ensured compliance with banking process controls, data accuracy, and audit requirements.

KEY ACHIEVEMENTS

- Automated 70% of regression workflows, reducing manual validation effort by 40%.
- Reduced recurring defects by 30% through improved process validation and automation coverage.
- Enabled business users through structured automation documentation and support.
- Led end-to-end testing for multiple banking automation releases with zero critical production defects.
- Successfully tested multiple Banking and Payment releases with zero critical production defects.
- Automated 70% of regression workflows reducing manual effort by 40% and improving release confidence.
- Improved automation execution reliability through reusable framework utilities and stable CI/CD-integrated pipelines.
- Successfully validated banking transaction workflows and enterprise releases with zero critical production defects.
- Played key role in production support ensuring zero critical release failures.
- Improved UAT efficiency by strengthening workflow-level validation strategies.
- Improved AI chatbot workflow validation coverage through enhanced intent, response, and fallback scenario testing.
- Reduced recurring production issues through proactive regression analysis, workflow-level validation, and backend verification strategies.
- Contributed to scalable automation framework improvements supporting faster execution and improved defect detection.
- Strengthened UAT and production validation processes through risk-based testing and business workflow verification.
- Played a key role in validating high-value payment transactions and reconciliation reports.

EDUCATION

- Bachelor of Engineering – Computer Science | Aksheyaa Engineering College, Anna University | 2012 – 2016

LANGUAGES : English – Fluent, Tamil – Native, Malayalam – Proficient